

Numerical Study of Concentration Gradient Generation in Y-Junction Microfluidic Devices

ME 224 PROJECT



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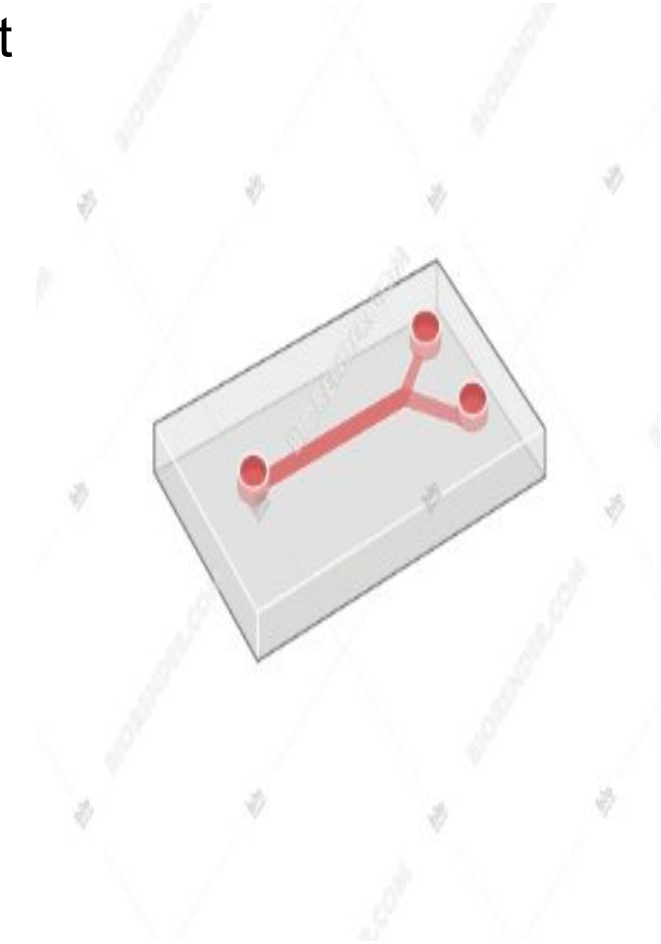
Madasi Benin

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Introduction

- In this project, two inlet streams carrying the same base fluid(water) but different solute(O₂) concentrations are introduced into a Y-junction device.
- Using numerical simulation (CFD), we will analyze how diffusion creates a concentration gradient across the channel width and height.
- In microfluidic systems, mixing occurs mainly through molecular diffusion, since turbulence is absent.
- Y-junction microchannels are commonly used to generate concentration gradients, where two inlet streams merge and interact.
- In microscale channels, flow is always laminar due to low reynolds number.



Literature Review

- Y-junction microchannels are widely used for diffusion-based gradient generation.
- CFD simulations are commonly used to model laminar flow and species transport.
- Gradient formation depends on Peclet number and inlet flow conditions.
- Channel geometry influences mixing behavior and gradient development.
- Limited studies compare junction angles under different concentration cases.

$$Re_D = \frac{\rho VD}{\mu} = \frac{VD}{\nu}$$

$$Pe = \frac{uL}{d}$$

Where:

u is the fluid velocity **L** is the characteristic length **D** is the diffusion coefficient

Motivation and objectives

Motivation

- Need for controlled gradient generation
- Importance in biomedical and chemical applications
- Need for optimization before fabrication

Objective

- To numerically simulate concentration gradient formation.
- To analyze effect of inlet concentration difference at different diffusion coefficient and different channel height and width.
- To study influence of junction angle.
- To evaluate gradient profiles using mass fraction plots.

Methodology

1. GEOMETRIC DESIGN & DIMENSIONS

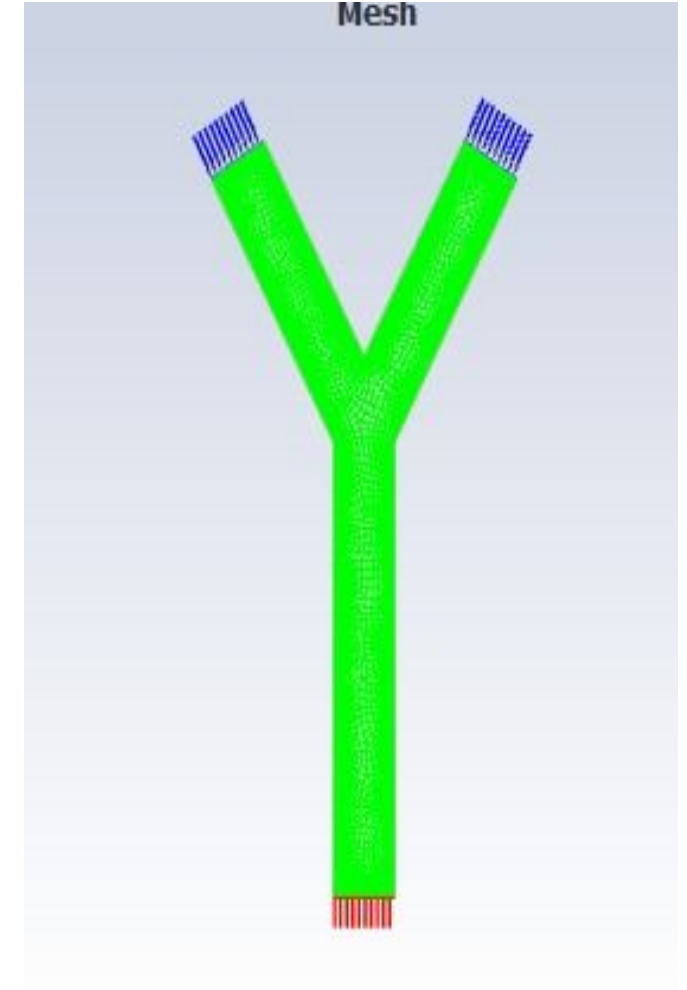
Inlet channel width = 25 microns

Inlet channel length = 100 microns

Main mixing channel width = 25 microns

Main mixing channel length = 150 microns

Angle between inlets = two cases: 90 and 60 degrees



2. GOVERNING EQUATIONS

Navier-Stokes Equation: Governs the velocity field of the fluid. Because of the micro-scale, viscous forces dominate over inertial forces.

Convection-Diffusion Equation: Since there is no turbulence, mixing occurs solely via molecular diffusion. The concentration changes across the channel width as the fluid moves downstream.

Simulation Setup: The Y-junction microchannel geometry(2D) is created using CAD software and is then imported to ANSYS for CFD simulation.

Concentration Analysis: We will simulate the interface where the two concentrations meet to measure the uniformity and stability of the resulting gradient.

Equations:- Navier-Stokes and Convection- Diffusion equation

$$\begin{array}{c}
 \text{Inertia (per volume)} \\
 \hline
 \underbrace{\frac{\partial \mathbf{u}}{\partial t}}_{\text{Variation}} + \underbrace{(\mathbf{u} \cdot \nabla) \mathbf{u}}_{\text{Convective acceleration}} = \underbrace{-\nabla w}_{\text{Internal source}} + \underbrace{\nu \nabla^2 \mathbf{u}}_{\text{Diffusion}} + \underbrace{\mathbf{g}}_{\text{External source}}
 \end{array}$$

$$\frac{\partial f}{\partial t} = D \frac{\partial^2 f}{\partial x^2} - \mu \frac{\partial f}{\partial x}$$

$$f(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{1}{4} \frac{(x - \mu t)^2}{Dt}}$$

3. **BOUNDARY CONDITIONS and PARAMETRIC CASES**

Diffusion coefficient (d1) = $1e-08$ m.m/s

Diffusion coefficient (d2) = $1e-011$ m.m/s

Both inlet initial velocity = 0.001 m/s

Viscosity = 0.001 kg/m.s

Concentration at inlets 1 and 2, C1 = (0 and 1)

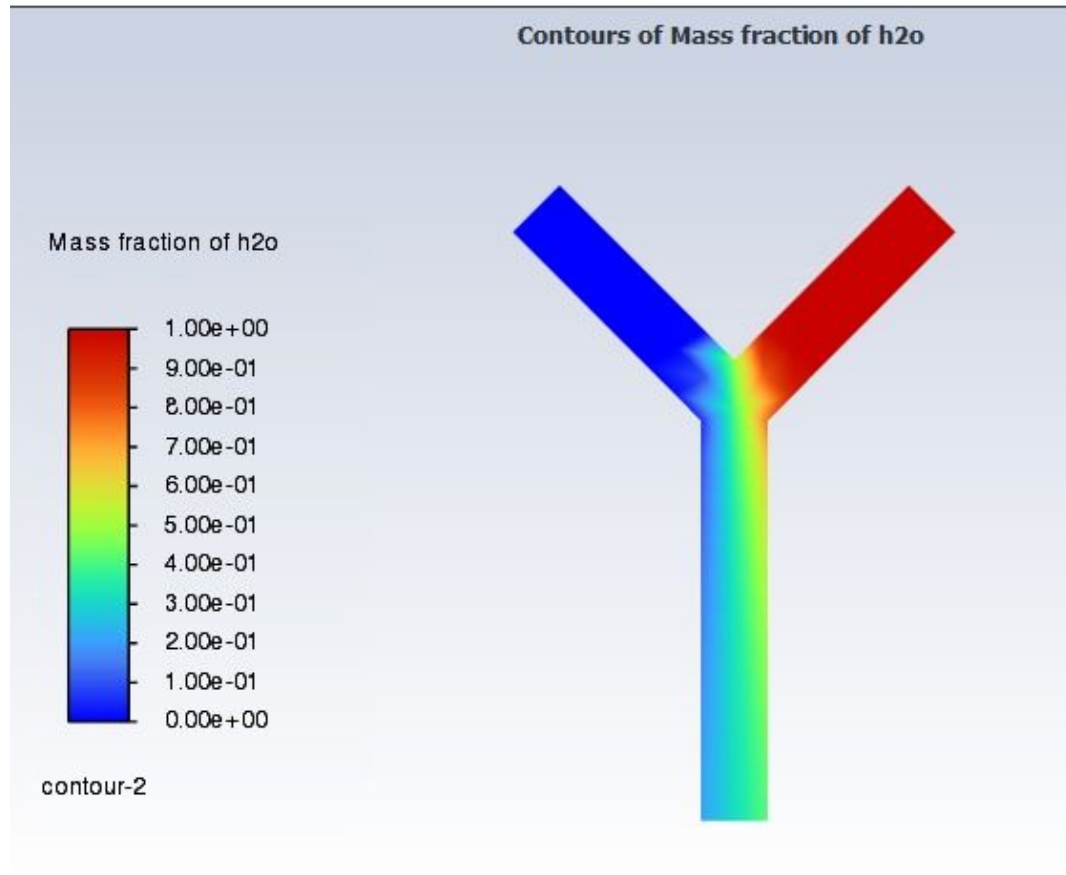
Concentration at inlets 1 and 2, C2 = (0.7 and 0.3)

Angle between inlets, A1 = 90 degrees

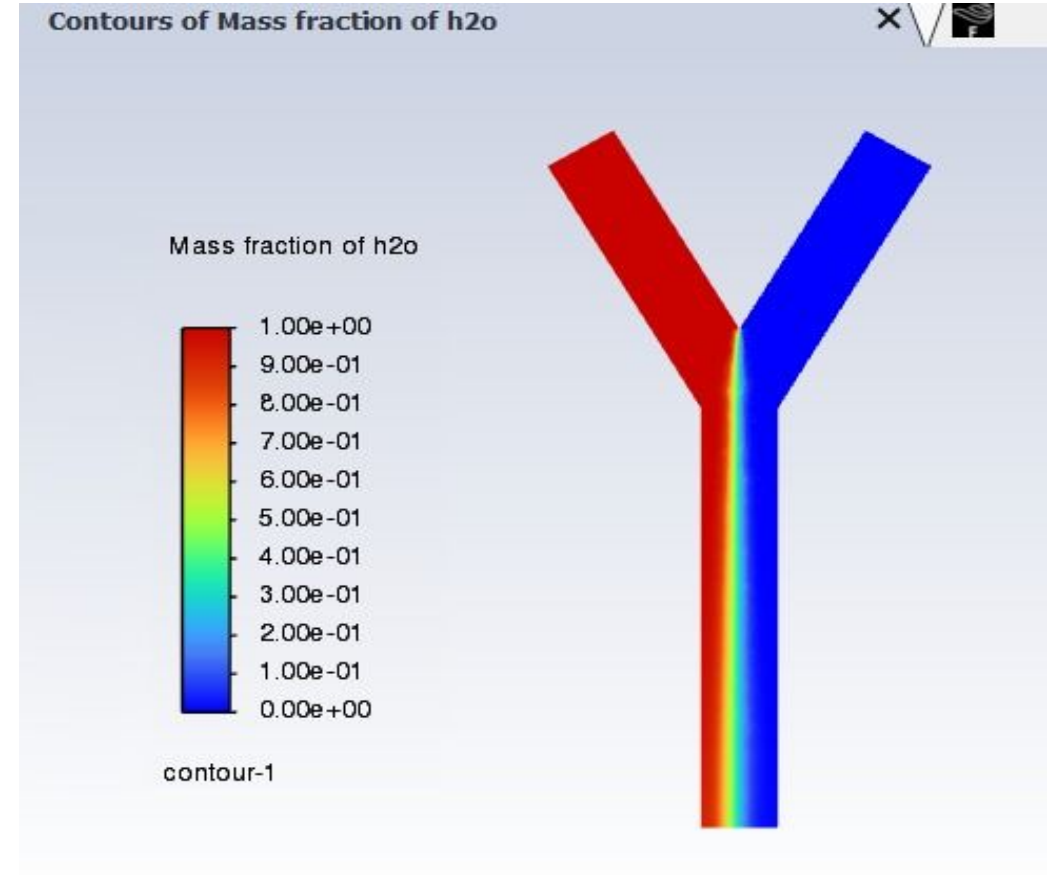
Angle between inlets, A2 = 60 degrees

Results and discussion

1. Gradient Formation in case of different angles and case C1(0,1):

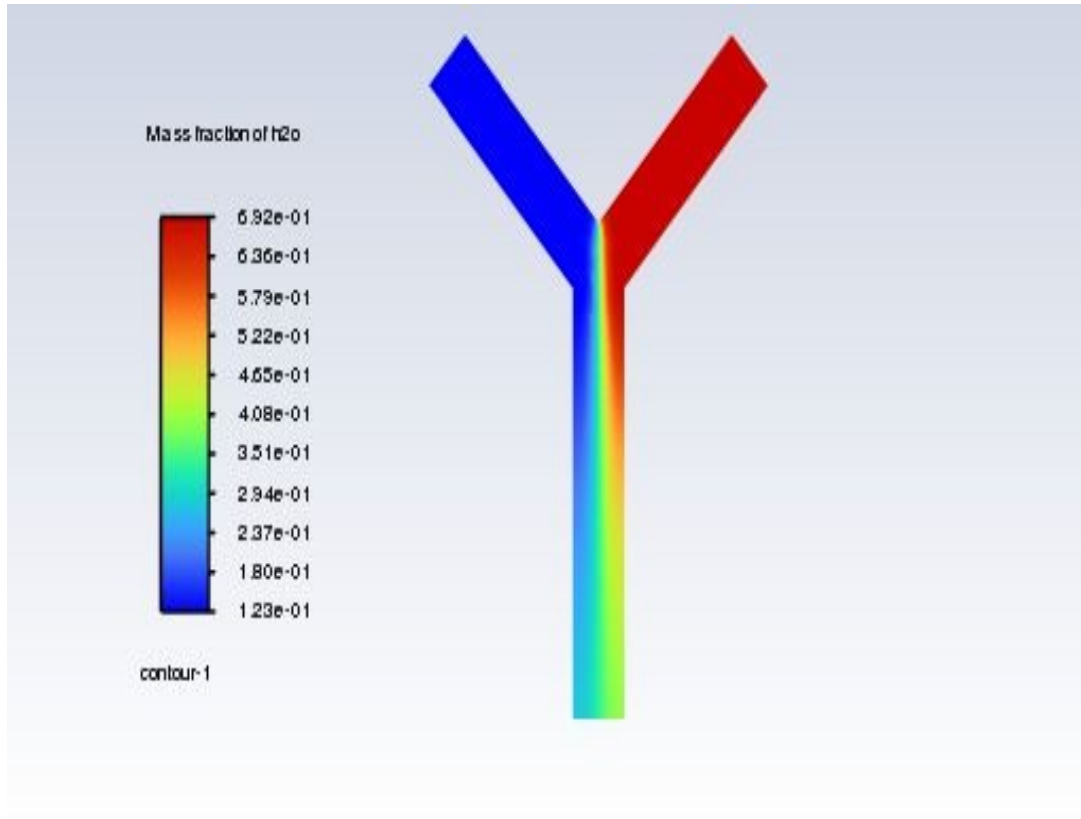


A1: 90 deg

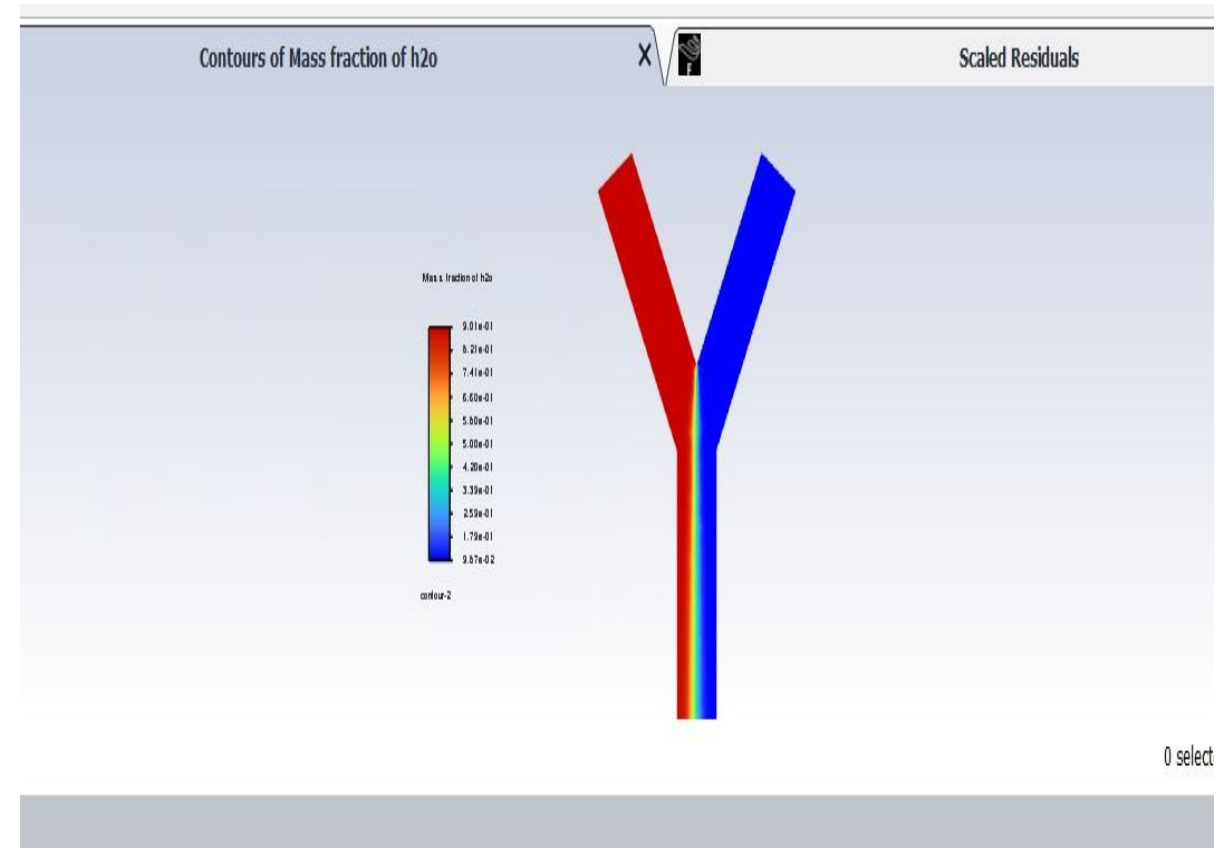


A2: 60 deg

Gradient formation in C2(0.7, 0.3):



A1: 90 deg



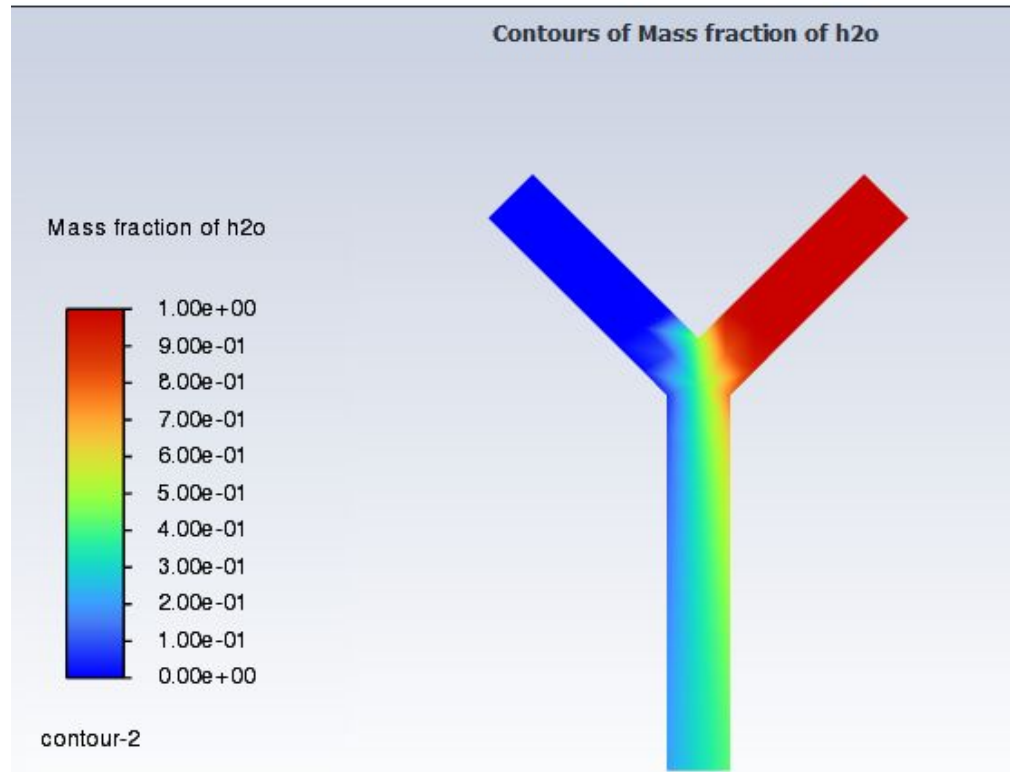
A2: 60 deg

What we analysed from this?

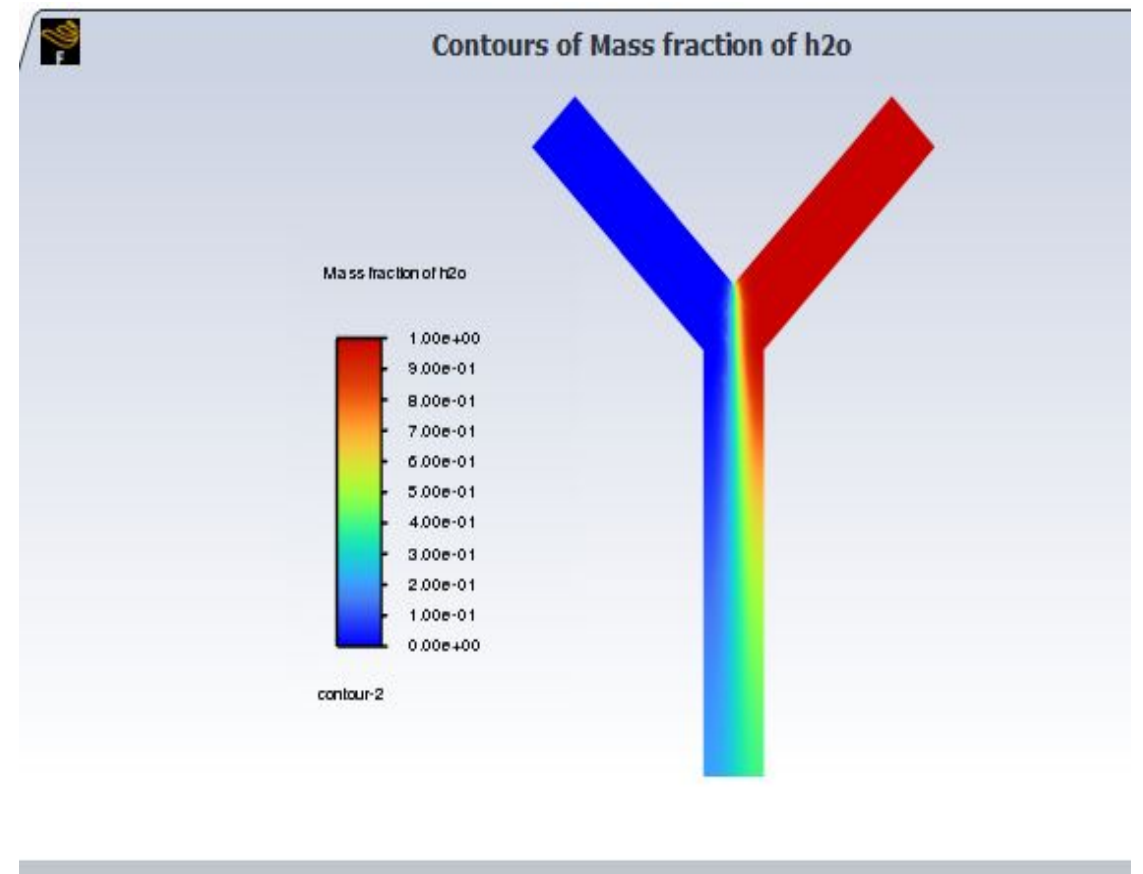
- Gradient formation is sharper and steeper when the inlet concentration difference is higher.
- The 90° junction produces a more symmetric and stable interface between the two streams due to equal merging direction, leading to smoother gradient development along the main channel.
- The 60° junction shows a slightly shifted and thinner interface, indicating enhanced convection effects and directional bias in gradient spreading.
- For smaller concentration difference (0.7–0.3), the gradient becomes smoother and less steep, as the driving diffusion force (concentration difference) is lower.

2. Comparison b/w diffusion coefficients for different angle

(a) Angle = 90 deg

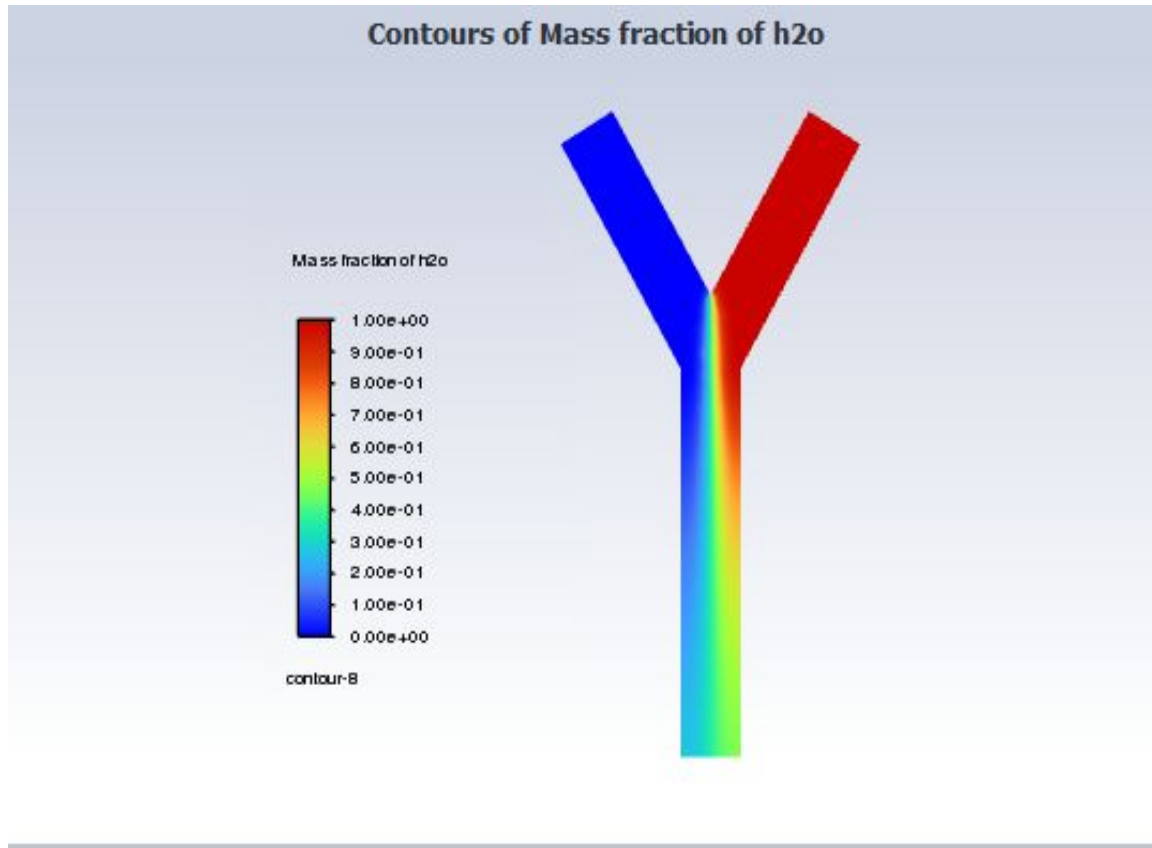


Diffusion coefficient= d_1

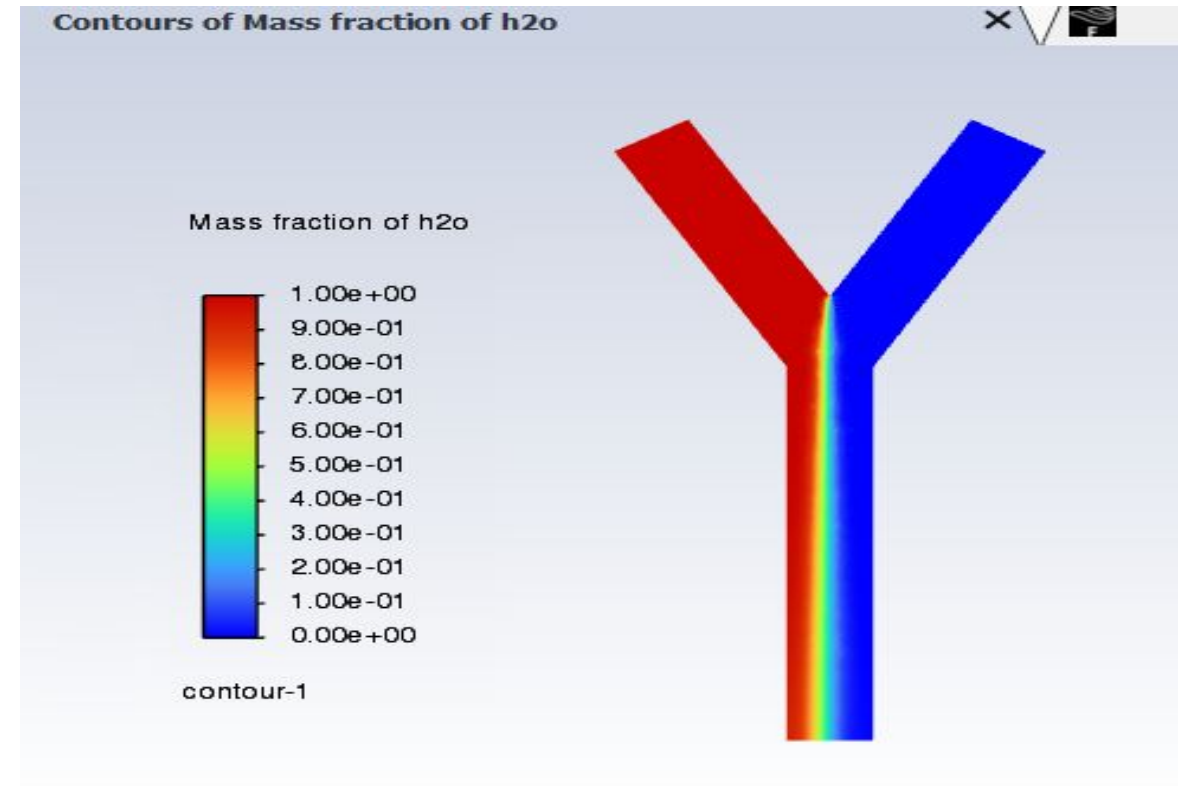


Diffusion coefficient= d_2

(b) Angle = 60 deg



Diffusion coefficient= d_1



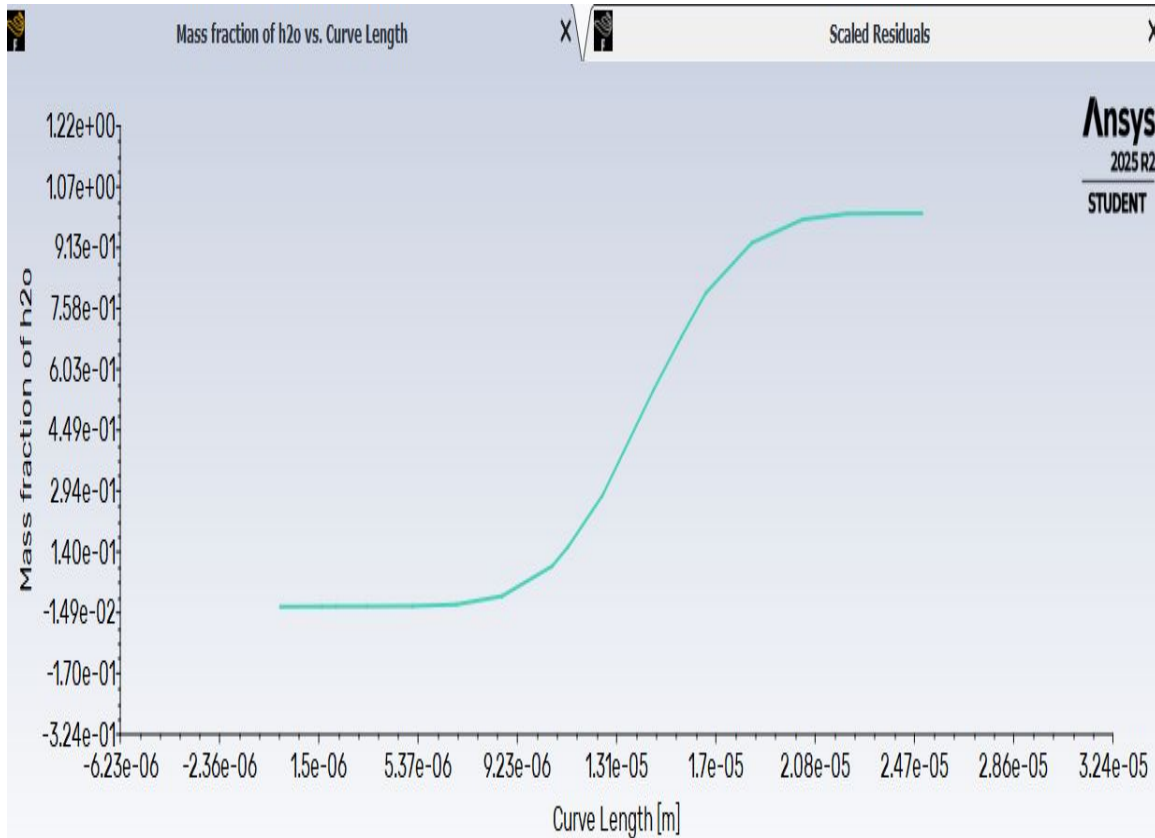
Diffusion coefficient= d_2

Comparison for different diffusion coefficients at different angles

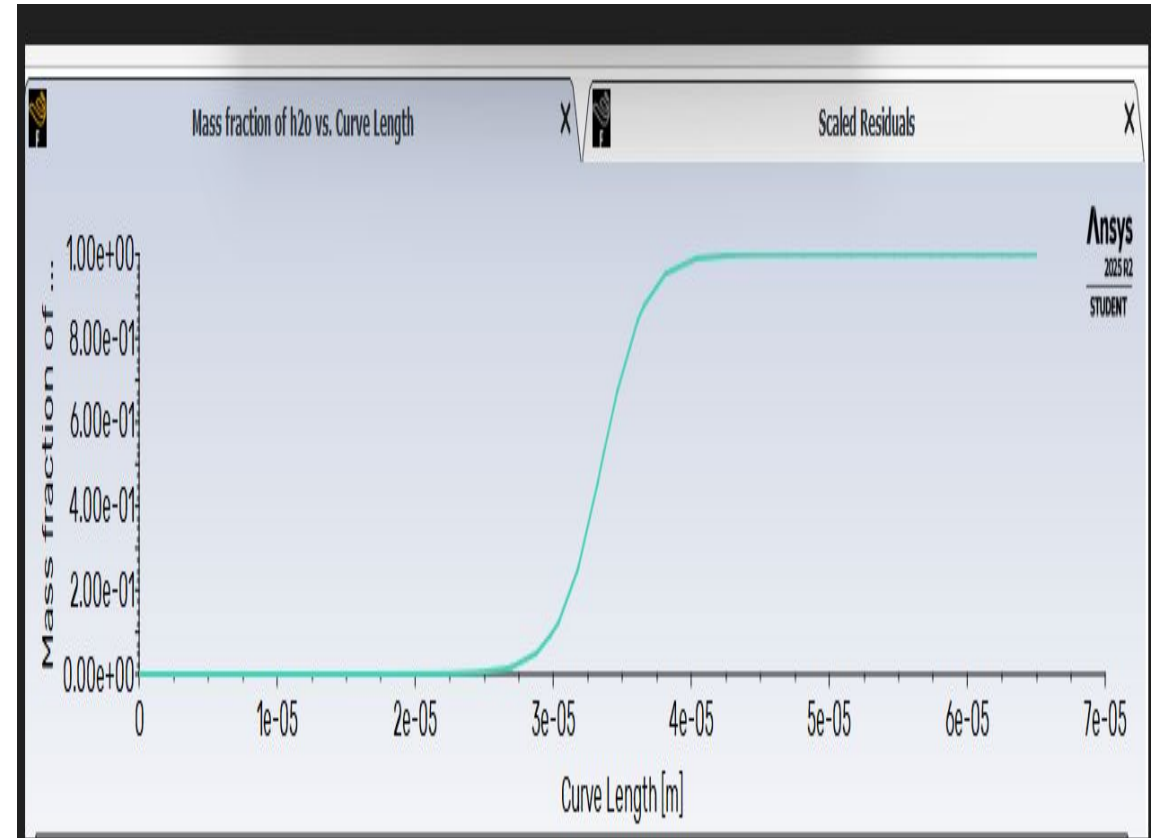
- Higher diffusion coefficient (D_2) produces a wider and smoother concentration gradient, indicating faster molecular diffusion between the two streams.
- Lower diffusion coefficient (D_1) results in a sharper and thinner interface, meaning convection dominates and mixing is weaker.
- For both 60° and 90° junctions, increasing diffusion coefficient increases gradient thickness and reduces concentration slope.
- The influence of diffusion coefficient is more visible near the junction region, where concentration difference is highest.

3. Gradient formation at different heights in main channel

(a) angle= 90 deg, near junction

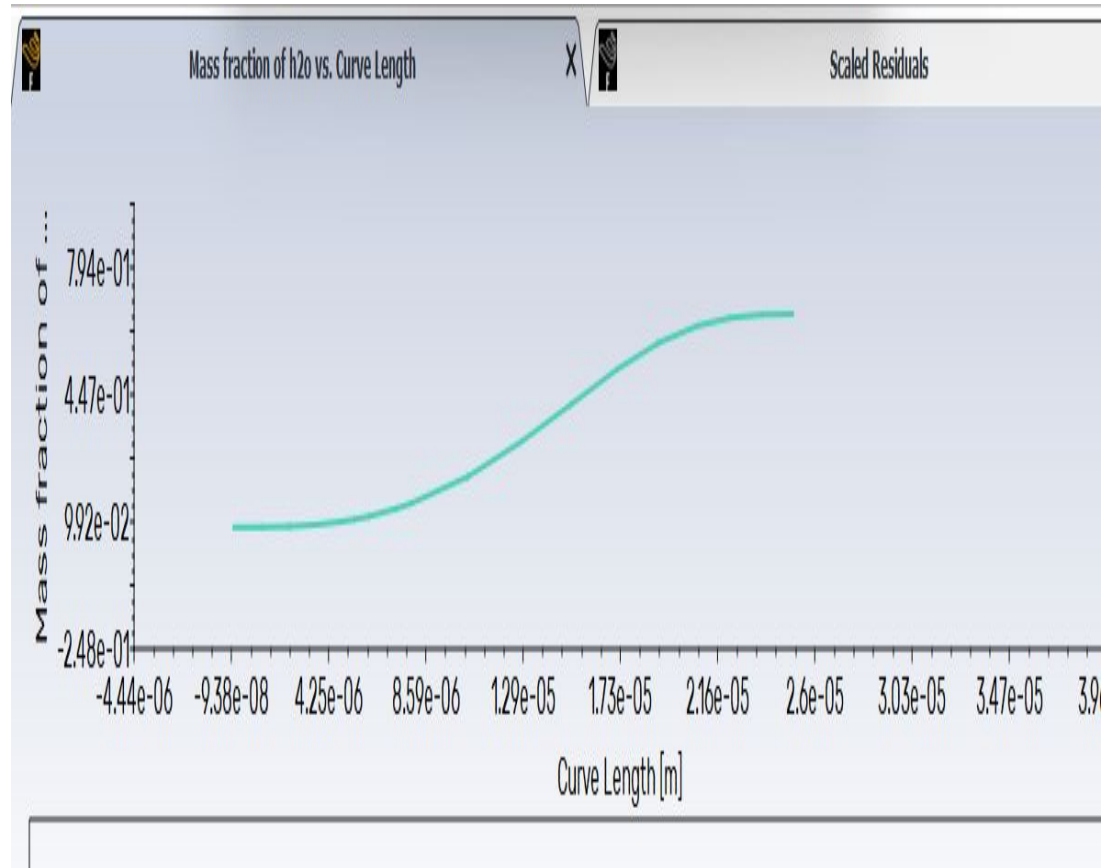


d1

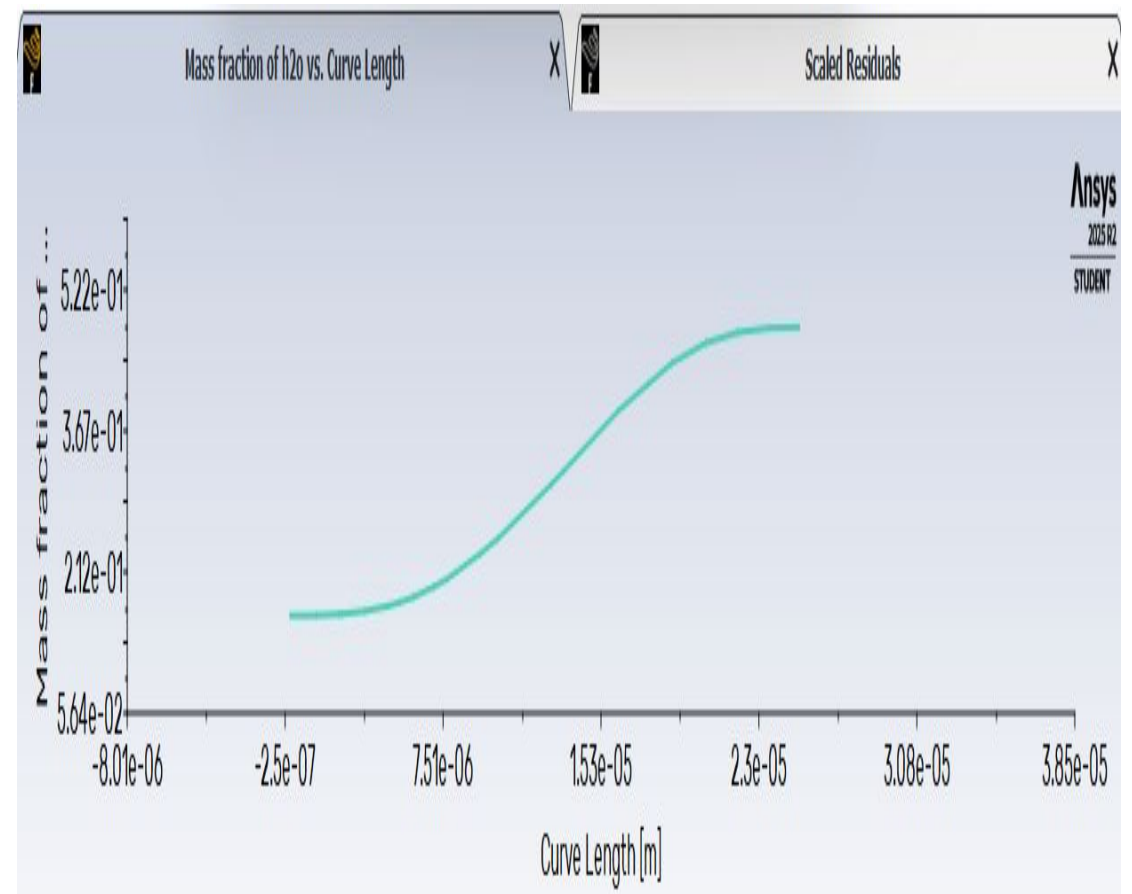


d2

(b) angle= 90 deg, 50 micron below junction

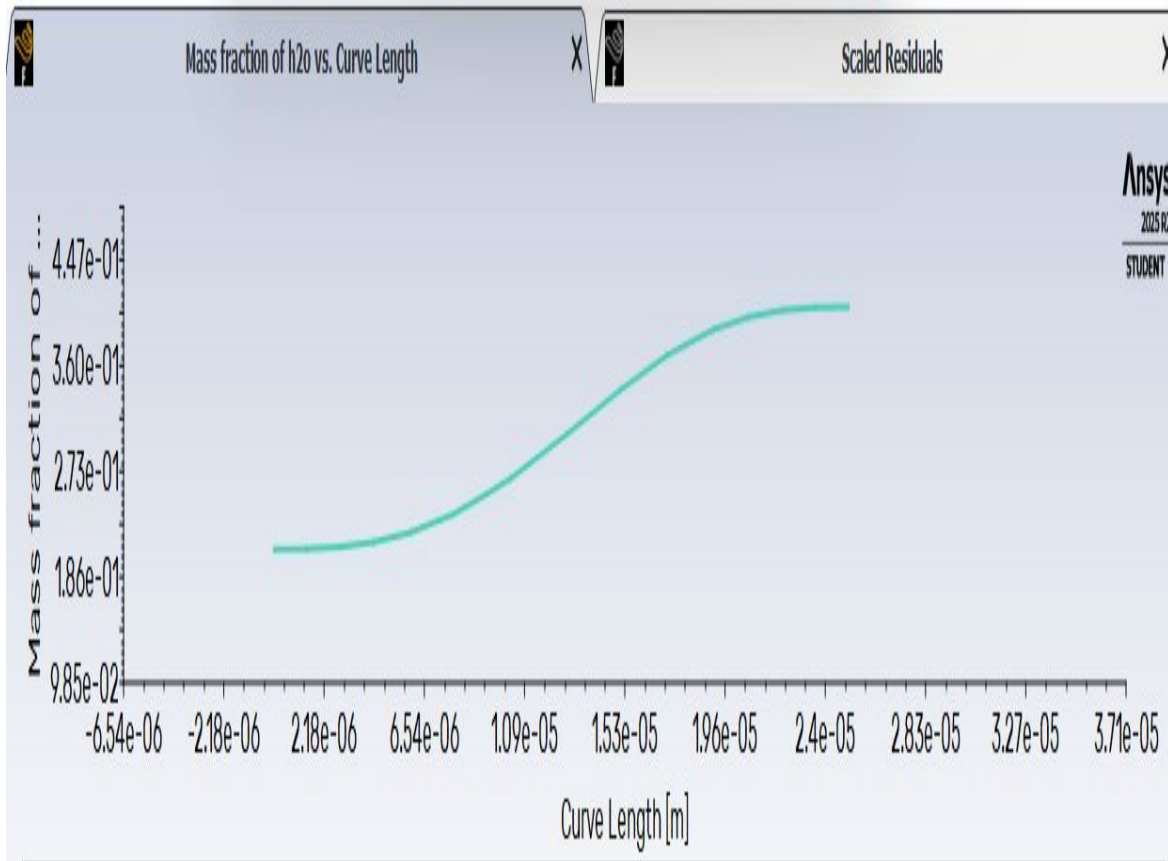


d1

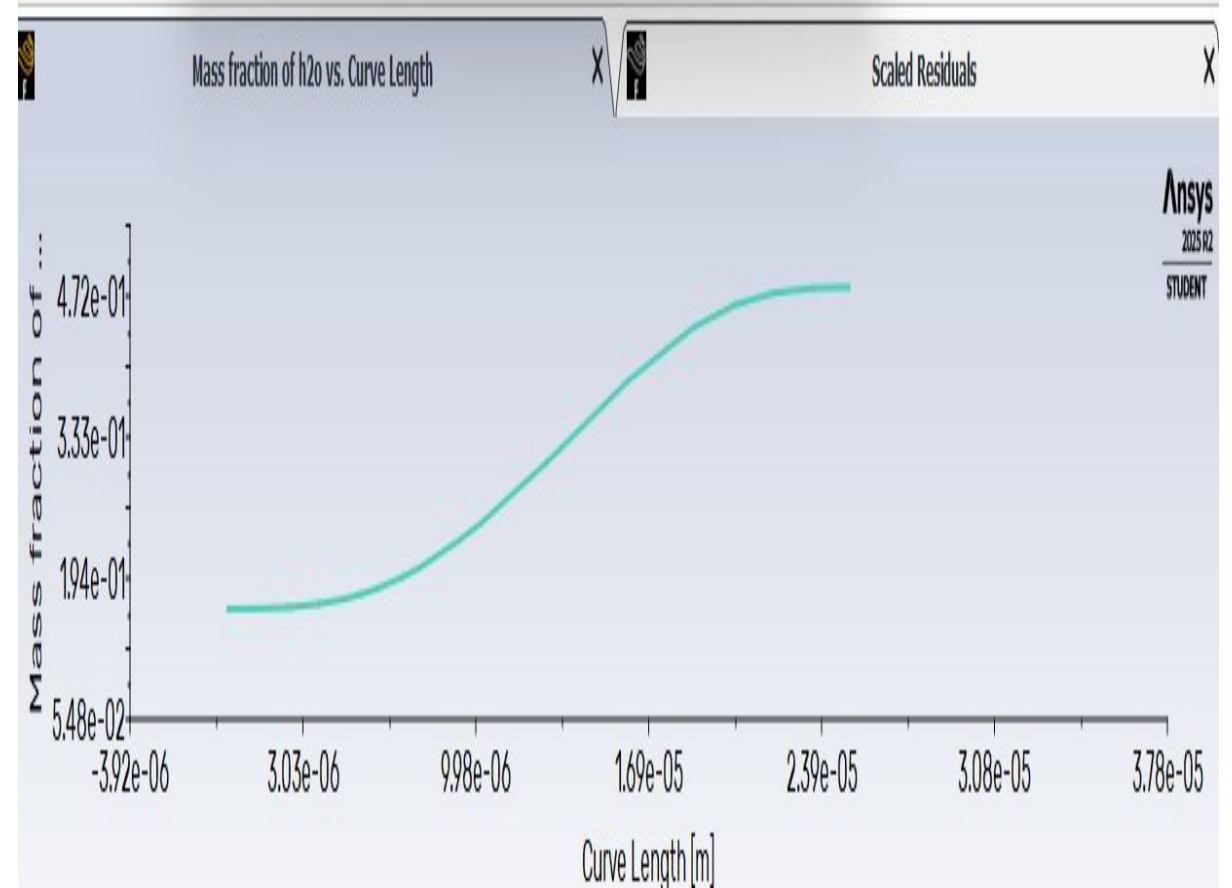


d2

(c) angle= 90 deg, 100 micron below junction

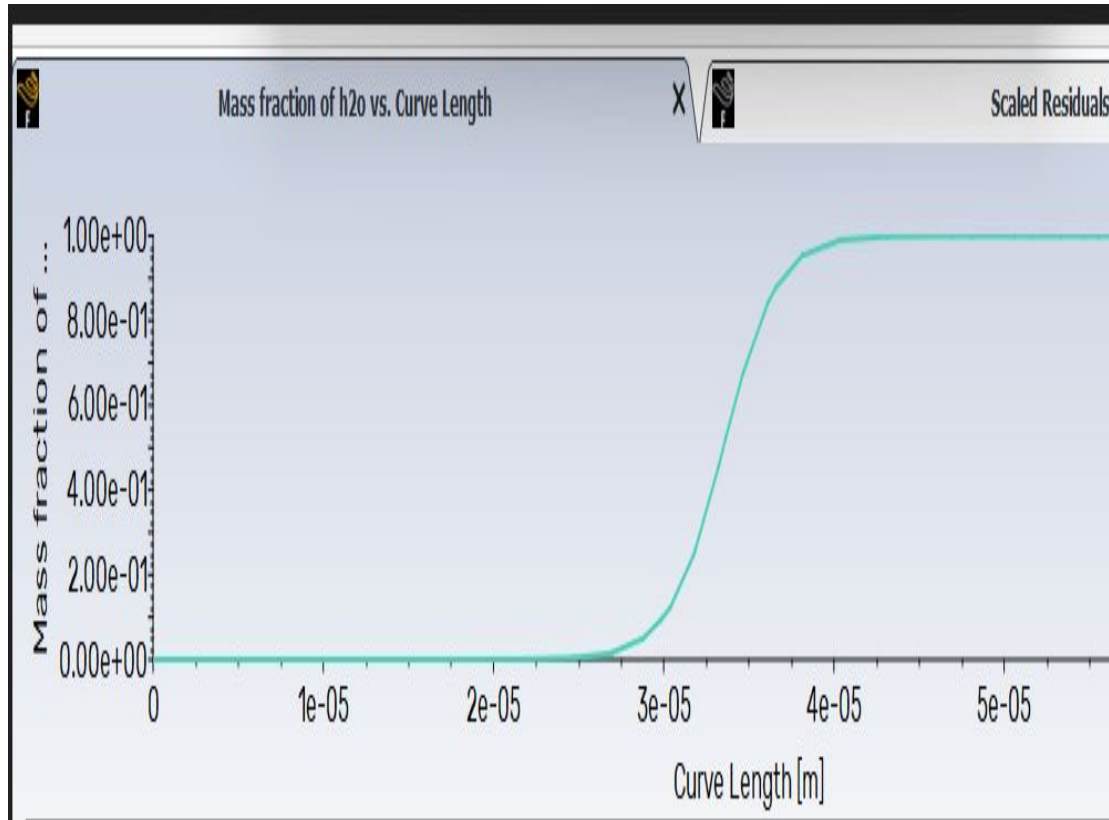


d1

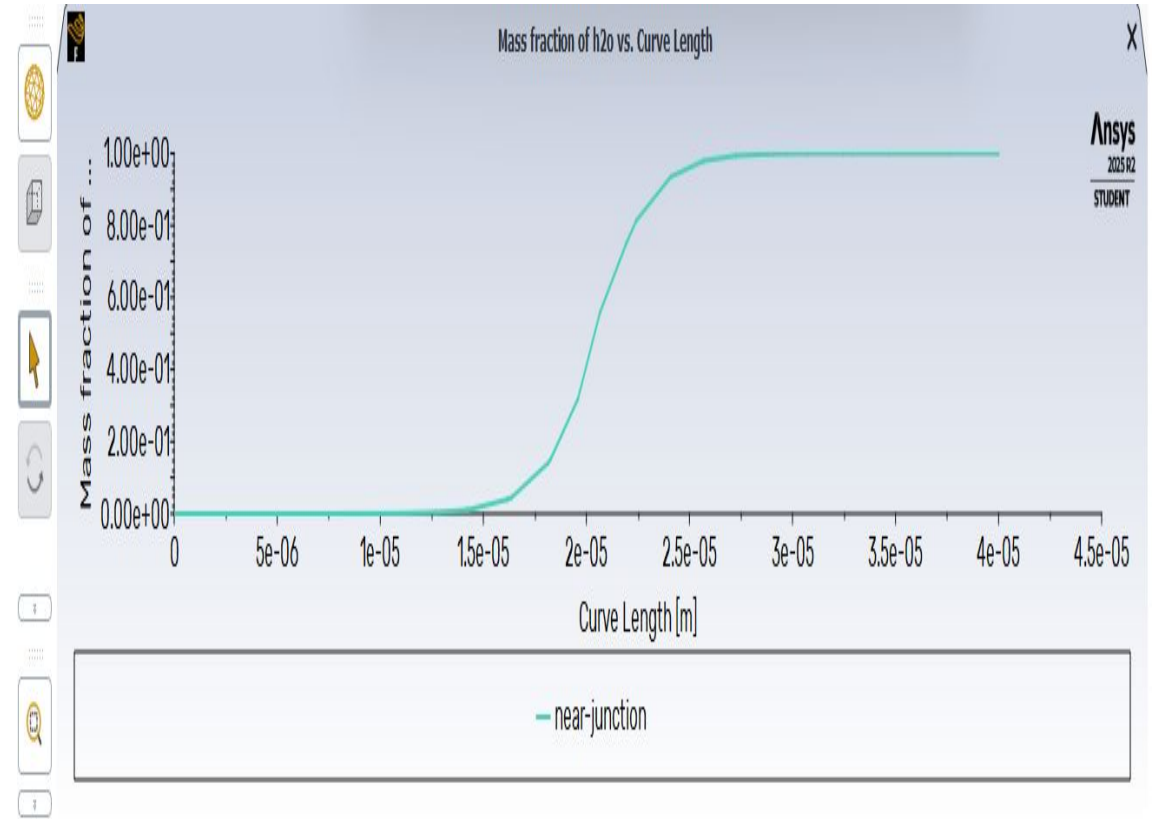


d2

(d)angle= 60 deg, near junction

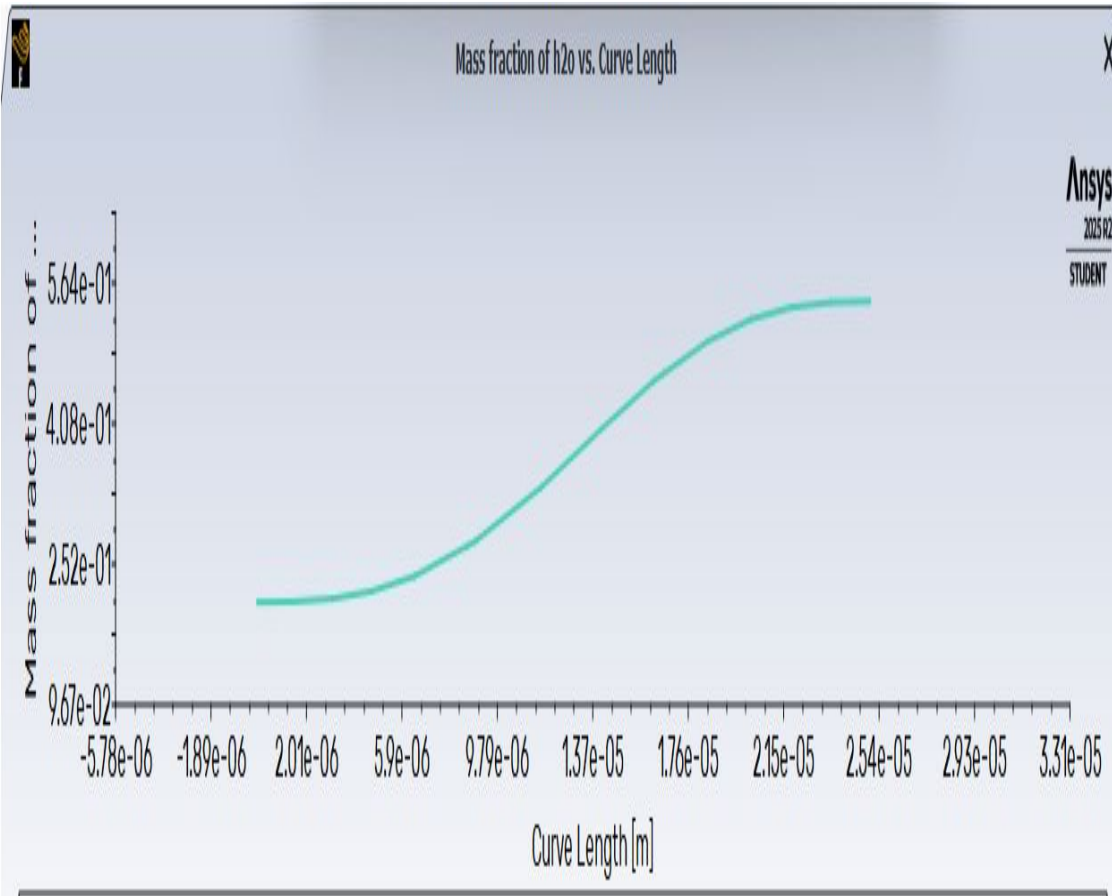


d1

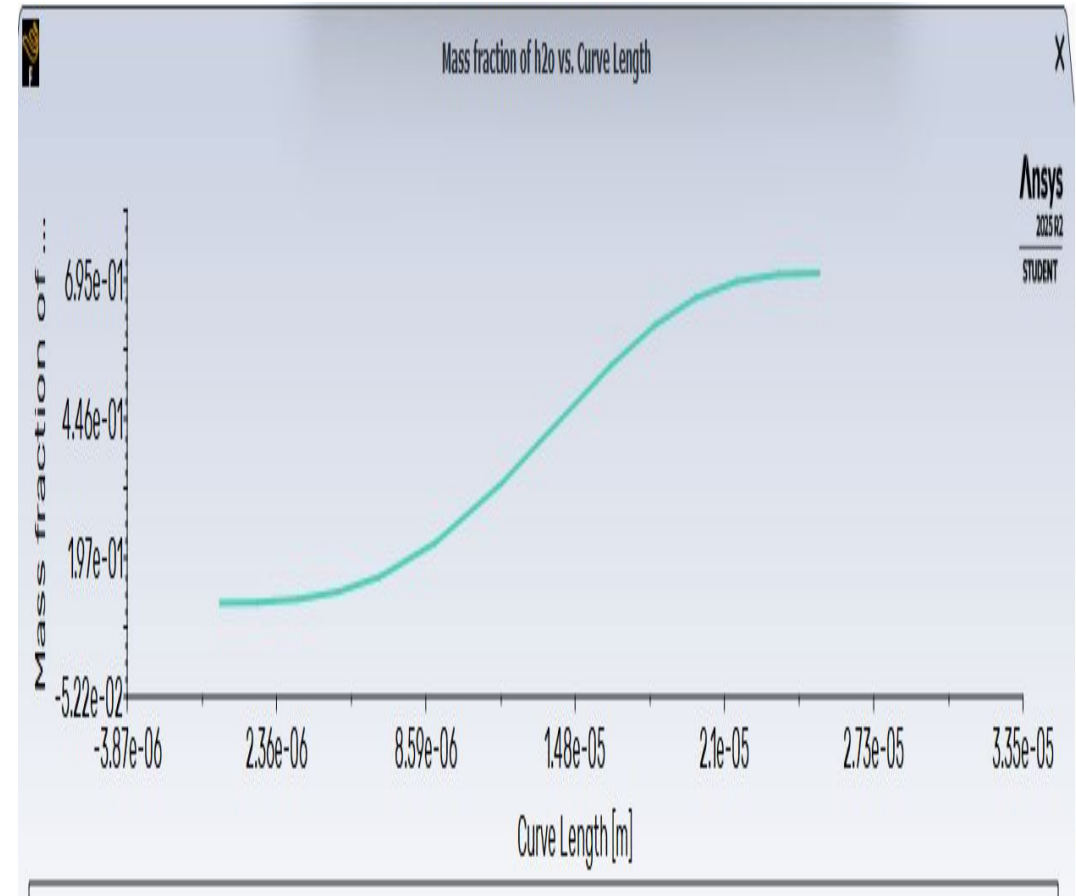


d2

(e)angle= 60 deg, 50 micron below junction

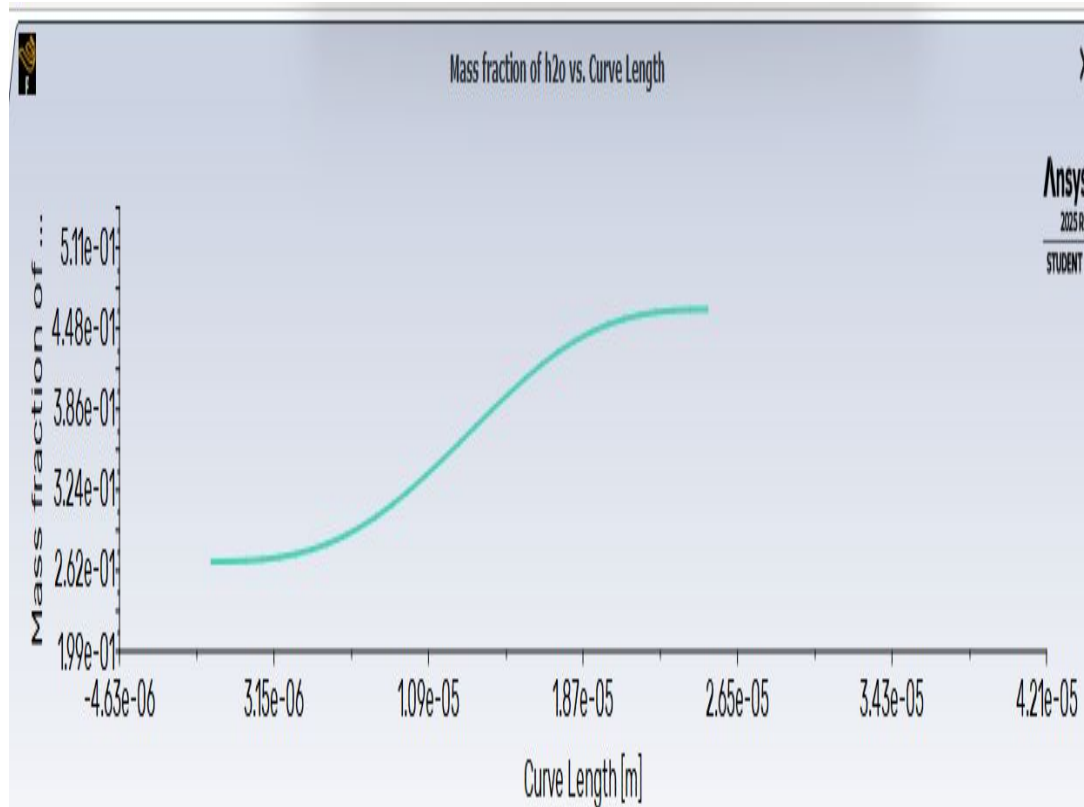


d1

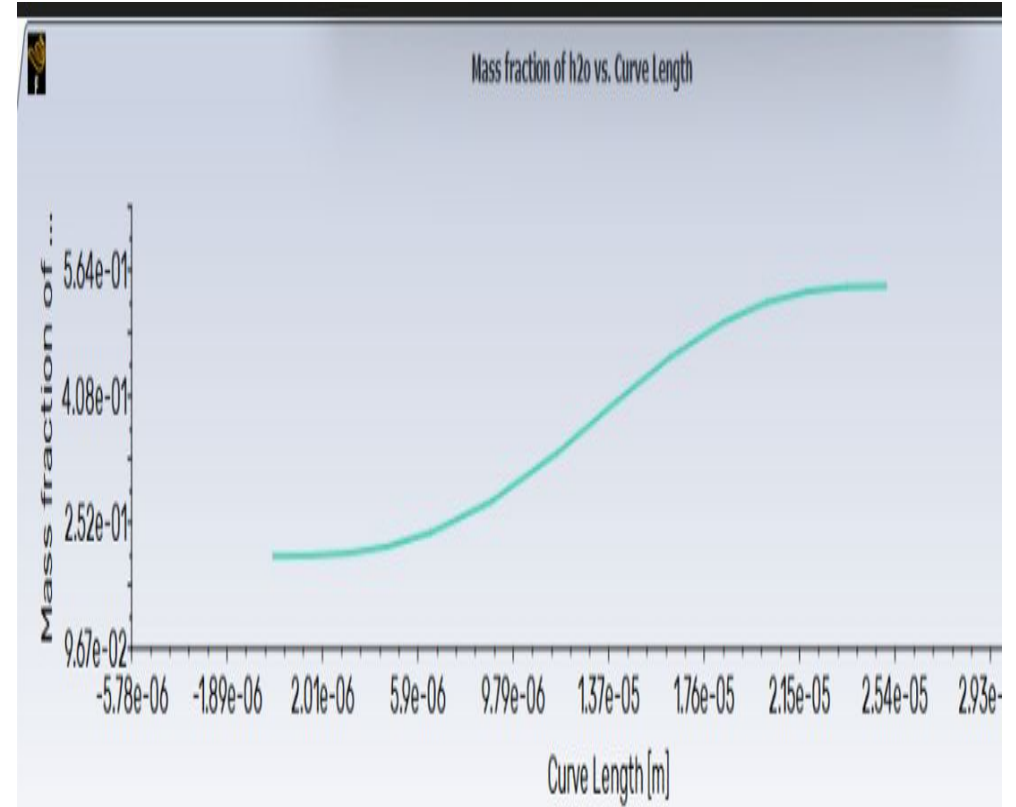


d2

(f) angle= 60 deg, near outlet



d1



d2

Gradient formation at different heights in the main channel

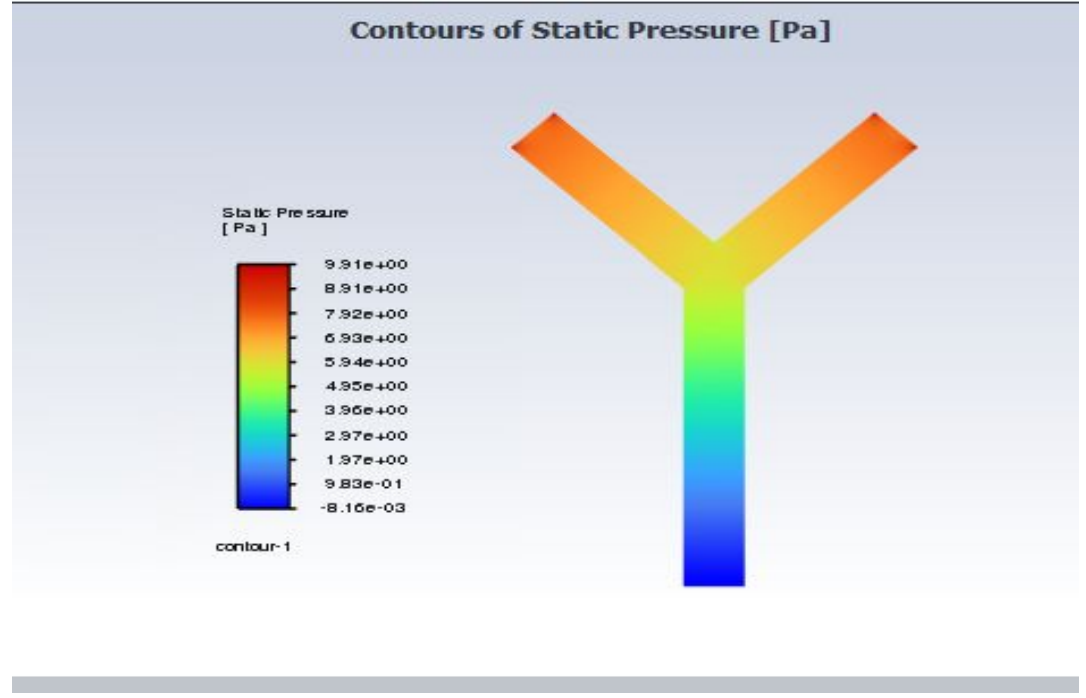
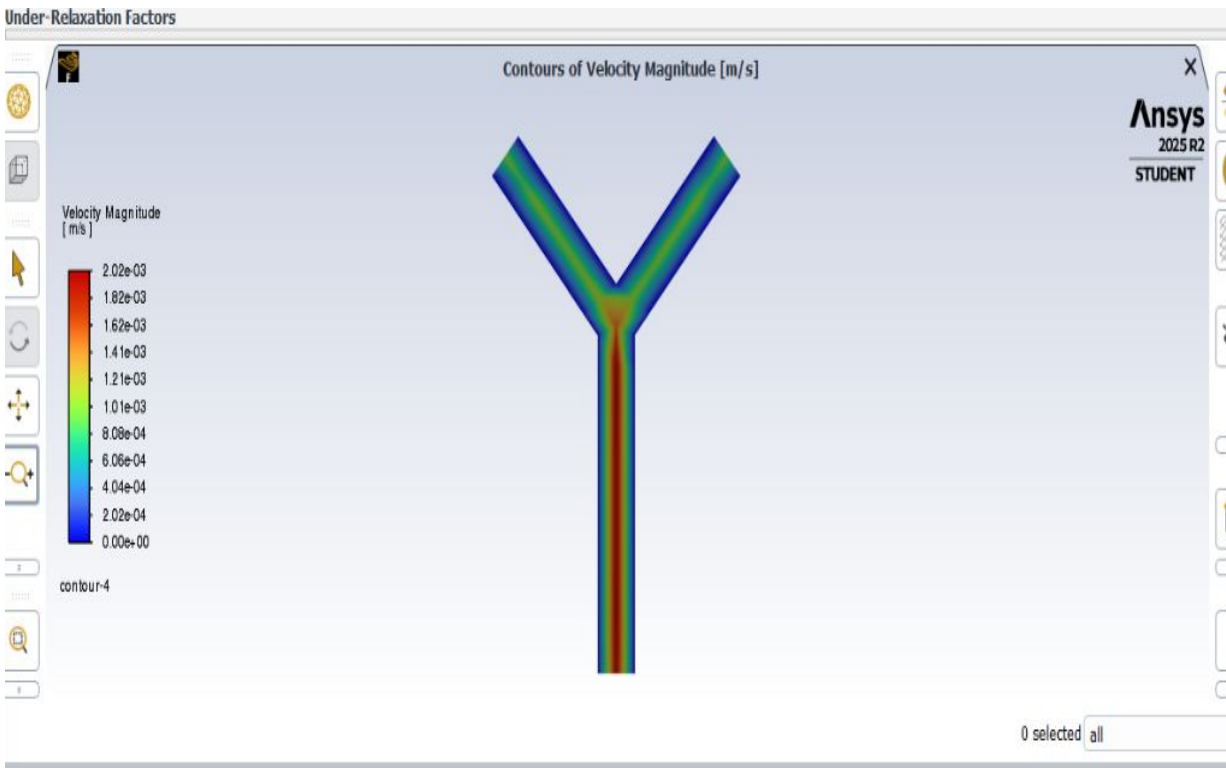
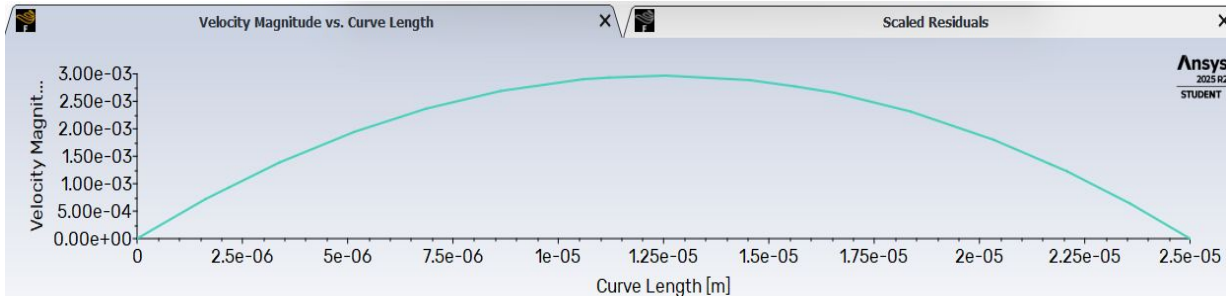
- At the junction (for both 60° and 90°), the concentration interface is sharp and narrow, since the two streams have just merged and diffusion time is minimal.
- At $50\text{ }\mu\text{m}$ below the junction, the gradient becomes wider and smoother due to increasing diffusion time along the channel length.
- Near the outlet, the concentration profile is more uniform, indicating progressive mixing as the fluid travels downstream.
- Comparing angles, the 90° junction produces a more symmetric gradient profile, while the 60° junction shows slight asymmetry and directional spreading due to flow alignment effects.

Calculation of Peclet and Reynolds Number

Case	D (m ² /s)	Re	Pe
1	1e-8	0.025	2.5
2	1e-11	0.025	2500

These are the values of peclet and reynolds number for different diffusion coefficients.

VELOCITY and PRESSURE CONTOUR



Conclusion on velocity and pressure distribution

- The velocity profile in the microchannel confirms laminar flow behavior, as indicated by the smooth and symmetric velocity contours.
- Velocity is higher in the center of the channel and lower near the walls due to the no-slip boundary condition, resulting in a parabolic-type profile.
- At the junction region, a slight velocity redistribution occurs due to merging of the two inlet streams, but no turbulence is observed.
- The pressure contour shows a gradual decrease from inlet to outlet, confirming steady pressure-driven flow in the microchannel.
- The smooth pressure drop and absence of fluctuations validate that the flow regime is stable and diffusion-dominated, which is essential for controlled gradient formation.

Overall Conclusion

In this numerical study of concentration gradient generation in a Y-junction microfluidic device, the effects of inlet concentration, diffusion coefficient, junction angle, downstream position, and Peclet number are analyzed under laminar flow conditions.

Case 1 – C1 (0, 1):

A sharp and strong concentration gradient is formed due to the large concentration difference.

Case 2 – C2 (0.7, 0.3):

A smoother and weaker gradient is observed because of the smaller diffusion driving force.

Diffusion Coefficient (d1 vs d2):

Higher diffusion coefficient results in faster and more uniform mixing, while lower diffusion maintains a distinct interface for a longer distance which also means that higher the Peclet number lower will be the diffusion and vice-versa.

Junction Angle (60° vs 90°):

The 60° configuration shows smoother stream interaction, whereas the 90° angle produces a more abrupt interface near the junction.

Peclet Number Effect:

Higher Peclet number (convection-dominated transport) preserves a sharper interface, while lower Péclet number enhances diffusion and promotes smoother gradient formation.

Downstream Effect:

As the flow moves downstream, the concentration profile gradually becomes smoother and more stable due to continuous molecular diffusion.

Overall, the study confirms that geometric parameters and transport properties critically control concentration gradient formation in microfluidic devices.

Future scope

- Study additional junction angles and geometric variations to identify the optimal configuration for stable and uniform gradient generation.
- Analyze the effect of different flow rates and Peclet numbers to understand convection–diffusion balance in more detail.
- Extend the current 2D simulation to a full 3D model for more realistic representation of microchannel behavior.
- Perform experimental validation by fabricating a Y-junction microfluidic device and comparing results with numerical simulations.
- Optimize the device for specific applications, such as drug delivery or cell migration studies, using numerical optimization techniques.

References

- https://www.researchgate.net/journal/Journal-of-Theoretical-and-Applied-Mechanics-2543-6309?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1YmxpY2F0aW9uliwicGFnZSI6InB1YmxpY2F0aW9uIn19
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